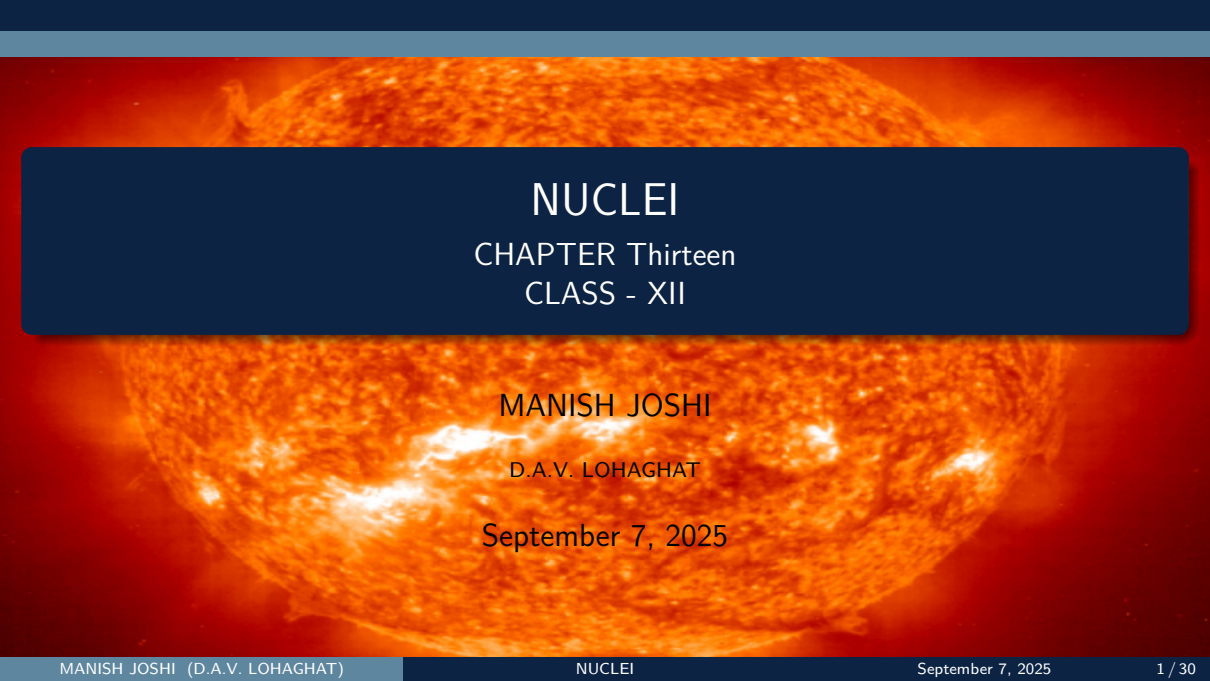




NUCLEI

CHAPTER Thirteen
CLASS - XII

MANISH JOSHI

D.A.V. LOHAGHAT

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Learning Objectives:

- Composition and size of nucleus, nuclear force
- Mass-energy relation, mass defect; binding energy per nucleon and its variation with mass number;
- Nuclear Fission, Nuclear Fusion.

Introduction

- In every atom, the **positive charge and mass** are **densely concentrated at the centre of the atom** forming its **nucleus**.
- the **radius of a nucleus** was smaller than the radius of an atom by a factor of about 10^4 .
- This means the **volume of a nucleus** is about 10^{-12} times the volume of the **atom**. In other words, an **atom is almost empty**.
- Nevertheless, the **nucleus contains most** (more than **99.9%**) of the **mass of an atom**.

ATOMIC MASSES AND COMPOSITION OF NUCLEUS

Atomic masses:

- The **mass of an atom is very small**, compared to a kilogram; for example, the **mass of a carbon atom, ^{12}C** , is **1.992647×10^{-26} kg**.
- **Kilogram is not a very convenient unit** to measure such small quantities.
- Therefore, **atomic mass unit (u)**, is **defined as 1/12th of the mass of the carbon (^{12}C) atom**. According to this definition

$$1u = \frac{\text{mass of one } ^{12}\text{C atom}}{12} = \frac{1.992647 \times 10^{-26}}{12} = 1.660539 \times 10^{-27} \text{ kg}$$

- **Accurate measurement of atomic masses** is carried out with a **mass spectrometer**.

ATOMIC MASSES AND COMPOSITION OF NUCLEUS

Atomic masses:

- mass of proton = mass of H atom - mass of electron
 $m_p = m_H - m_e = 1.00783 \text{ u} - 0.00055 \text{ u} = 1.00727 \text{ u} = 1.67262 \times 10^{-27} \text{ kg}$
- It was earlier thought that the nucleus may contain electrons, but this was ruled out later using arguments based on quantum theory. **All the electrons of an atom are outside the nucleus.**
- The **number of protons** in the nucleus of the atom is **Z**, the **atomic number**.
- The **charge of the nucleus** is **(+Ze)**.
- since the atom is neutral total **charge of the atomic electrons** is thus **(-Ze)**

ATOMIC MASSES AND COMPOSITION OF NUCLEUS

Discovery of Neutron:

- Since the **nuclei of deuterium and tritium are isotopes of hydrogen**, they must **contain only one proton** each. But the **masses of the nuclei of hydrogen, deuterium and tritium** are in the **ratio of 1:2:3**. Therefore, the nuclei of deuterium and tritium **must contain**, in addition to a proton, some **neutral matter**.
- This hypothesis was verified in 1932 by **James Chadwick** who **observed emission of neutral radiation** when **beryllium nuclei were bombarded with alpha-particles**

ATOMIC MASSES AND COMPOSITION OF NUCLEUS

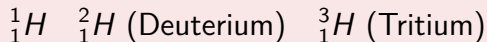
Discovery of Neutron:

- Chadwick assume that the neutral radiation consists of a new type of neutral particles called **neutrons**. From conservation of energy and momentum, he was able to determine the **mass of new particle as very nearly the same as mass of proton**
- mass of neutron $m_n = 1.00866u = 1.6749 \times 10^{-27} \text{ kg}$

ATOMIC MASSES AND COMPOSITION OF NUCLEUS

ISOTOPES

- **Isotopes:** Atoms of same element (same Z) but different A .
- Example: Hydrogen has three isotopes:



- Chlorine has isotopes $34.98u$ (75.4%) and $36.98u$ (24.6%). Average atomic mass $= 35.47u$.
- Chemical properties of isotopes are identical (same electronic configuration).

ATOMIC MASSES AND COMPOSITION OF NUCLEUS

COMPOSITION OF NUCLEUS

- The composition of a nucleus can now be described using the following terms and symbols:
- **Z = atomic number = number of protons**
- **N = number of neutrons**
- **A = mass number = Z + N = total no. of proton and neutron (or nucleons)**
- Nuclear species or nuclides are shown by the notation A_ZX where X is the chemical symbol of the species.
- For example, the nucleus of gold is denoted by ${}^{197}_{79}Au$. It contains 197 nucleons, of which 79 are protons and the rest 118 are neutrons.

ATOMIC MASSES AND COMPOSITION OF NUCLEUS

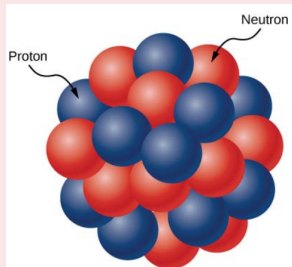
ISOBARS AND ISOTONES

- **Isobars:** Same mass number (A), different Z . Example: ${}^3_1\text{H}$ and ${}^3_2\text{He}$.
- **Isotones:** Same neutron number (N), different Z . Example: ${}^{198}_{80}\text{Hg}$ and ${}^{197}_{79}\text{Au}$.

SIZE OF THE NUCLEUS

- Rutherford's suggestion, Geiger and Marsden performed their classic experiment: on the scattering of α -particles from thin gold foils. Their experiments revealed that the distance of closest approach to a gold nucleus of an α -particle of kinetic energy 5.5 MeV is about 4.0×10^{-14} m.
- the **actual size of the nucleus has to be less than 4.0×10^{-14} m.**
- By performing scattering experiments in which **fast electrons**, instead of α -particles, are projectiles that bombard targets made up of various elements, the **sizes of nuclei of various elements have been accurately measured.**

SIZE OF THE NUCLEUS



- The simplest model of the nucleus is a densely packed sphere of nucleons. **The volume V of the nucleus** is therefore **proportional to the number of nucleons A** , expressed by

$$V = \frac{4}{3}\pi R^3 = kA$$

- where **R is the radius of a nucleus** and **k is a constant** with units of volume. Solving for R , we have

$$R = R_0 A^{1/3}$$

- where **$R_0 = 1.2 \times 10^{-15} \text{ m} (= 1.2 \text{ fm})$** .

DENSITY OF THE NUCLEUS

- $\text{Density} = \frac{\text{Mass of nucleus}}{\text{Volume of nucleus}} = \frac{A}{\frac{4}{3}\pi R^3} = \frac{A}{\frac{4}{3}\pi R_0^3 A}$
- $\text{Density} = \frac{1}{\frac{4}{3}\pi R_0^3} = \text{Constant} = 2.3 \times 10^{17} \text{ kgm}^{-3}$

Example 1

Given the **mass** of iron nucleus as **55.85 u** and **A=56**, find the **nuclear density**?
($1u = 9.27 \times 10^{-26} \text{ kg}$)

Mass – Energy

- **Einstein** showed from his **theory of special relativity** that it is necessary to **treat mass as another form of energy**.
- **Before** the advent of this **theory of special relativity** it was **presumed that mass and energy were conserved separately** in a reaction.
- However, **Einstein showed** that **mass is another form of energy** and **one can convert mass-energy into other forms of energy**, say kinetic energy and vice-versa.
- Einstein gave the famous **mass-energy equivalence relation**

$$E = mc^2$$

Here, m is energy equivalent of mass and c is speed of light in vacuum.

Example 2

Calculate the energy equivalent of 1 g of substance.

Mass Defect

- We have seen that the **nucleus is made up of neutrons and protons.**
- Therefore **it may be expected** that the **mass of the nucleus is equal to the total mass of its individual protons and neutrons.**
- However, According to nuclear particle experiments, the **total mass of a nucleus (M_{nuc}) is less than the sum of the masses of its constituent nucleons (protons and neutrons).**

Mass Defect

- For example, let us consider ${}^{16}_8\text{O}$; a nucleus which has **8 neutrons and 8 protons**.
- Mass of 8 neutrons $= 8 \times 1.00866 \text{ u}$
- Mass of 8 protons $= 8 \times 1.00727 \text{ u}$
- Mass of 8 electrons $= 8 \times 0.00055 \text{ u}$
- Therefore the **expected mass** ${}^{16}_8\text{O}$ **nucleus** $= 8 \times 2.01593 \text{ u} = 16.12744 \text{ u}$.
- The **atomic mass** of ${}^{16}_8\text{O}$ **found from mass spectroscopy** experiments is seen to be **15.99493 u**. **Subtracting the mass of 8 electrons** ($8 \times 0.00055 \text{ u}$) from this, we get the **experimental mass** of ${}^{16}_8\text{O}$ **nucleus** to be **15.99053 u**.
- Thus, we find that the **mass of the** ${}^{16}_8\text{O}$ **nucleus is less than the total mass of its constituents by 0.13691 u**.

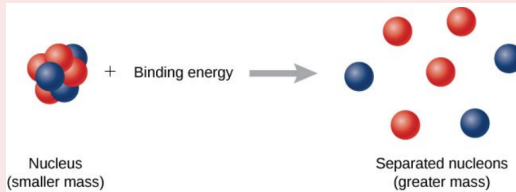
Mass Defect

- The **difference in mass of a nucleus and its constituents**, ΔM , is called the **mass defect**, and is given by

$$\Delta M = [ZM_p + (A - Z)M_n] - M_{nucl}$$

- where ZM_p is the **total mass of the protons**, $(A - Z)M_n$ is the **total mass of the neutrons**, and M_{nuc} is the **mass of the nucleus**.

Binding Energy



- The **formation of a nucleus** from a system of isolated protons and neutrons is therefore an **exothermic reaction**-meaning that it **releases energy**. The **energy emitted**, or radiated, in this process is $(\Delta M)c^2$.
- Now **imagine this process occurs in reverse**. Instead of forming a nucleus, **energy is put into the system to break apart the nucleus**. The amount of energy required is called the **total binding energy (BE)**, $(Eb = \Delta Mc^2)$.

Example 13.3 Find the energy equivalent of one atomic mass unit, first in Joules and then in MeV. Using this, express the mass defect of $^{16}_8\text{O}$ in MeV/c^2 .

Solution

$$1\text{u} = 1.6605 \times 10^{-27} \text{ kg}$$

To convert it into energy units, we multiply it by c^2 and find that energy equivalent = $1.6605 \times 10^{-27} \times (2.9979 \times 10^8)^2 \text{ kg m}^2/\text{s}^2$

$$= 1.4924 \times 10^{-10} \text{ J}$$

$$= \frac{1.4924 \times 10^{-10}}{1.602 \times 10^{-19}} \text{ eV}$$

$$= 0.9315 \times 10^9 \text{ eV}$$

$$= 931.5 \text{ MeV}$$

$$\text{or, } 1\text{u} = 931.5 \text{ MeV}/c^2$$

$$\begin{aligned} \text{For } ^{16}_8\text{O}, \quad \Delta M &= 0.13691 \text{ u} = 0.13691 \times 931.5 \text{ MeV}/c^2 \\ &= 127.5 \text{ MeV}/c^2 \end{aligned}$$

The energy needed to separate $^{16}_8\text{O}$ into its constituents is thus $127.5 \text{ MeV}/c^2$.

Graph of Binding Energy per Nucleon

- In nuclear physics, one of the most important experimental quantities is the binding energy per nucleon (E_{bn}), which is the **ratio of the binding energy** E_b of a nucleus **to the number of the nucleons**, A , in that nucleus:

$$E_{bn} = \frac{E_b}{A}$$

- We can think of **Binding energy per nucleon** as the **average energy per nucleon needed to separate a nucleus into its individual nucleons**.
- If the **Binding energy per nucleon** is relatively **large**, the **nucleus is relatively stable**.

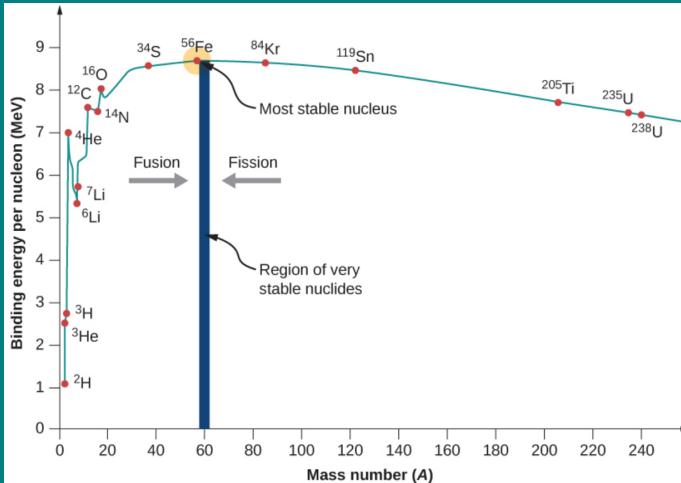
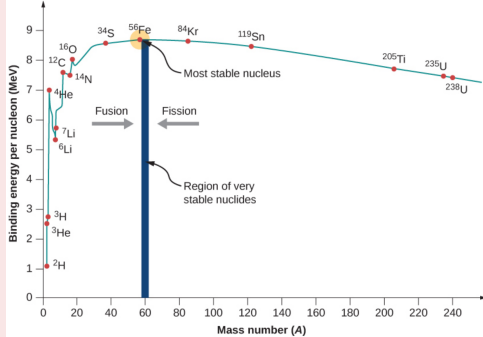


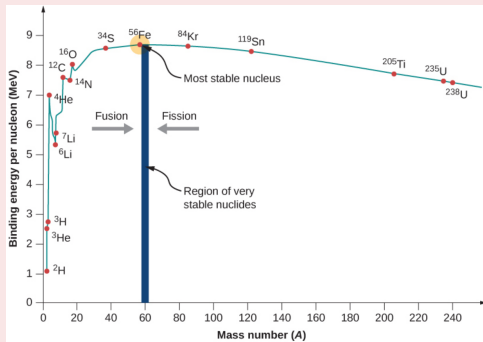
Figure 10.7 In this graph of binding energy per nucleon for stable nuclei, the BEN is greatest for nuclei with a mass near ^{56}Fe . Therefore, fusion of nuclei with mass numbers much less than that of Fe, and fission of nuclei with mass numbers greater than that of Fe, are exothermic processes.

Graph of Binding Energy per Nucleon



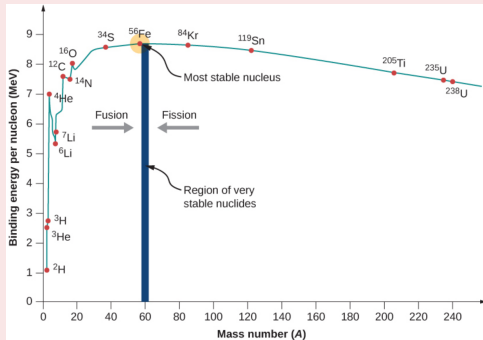
- We notice the following **main features of the plot**:
- the **binding energy per nucleon**, E_{bn} , is practically **constant**, i.e. practically independent of the atomic number of middle mass number nuclei ($30 < A < 170$).
- The curve has a **maximum** of about 8.75 MeV **for Fe** $A = 56$ and has a value of 7.6 MeV
- E_{bn} is **lower** for both **light nuclei** ($A < 30$) and **heavy nuclei** ($A > 170$).

Graph of Binding Energy per Nucleon



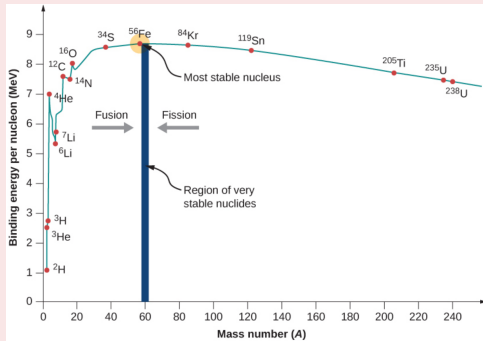
- typical E_{bn} values range from 6-10 MeV, with an average value of about 8 MeV.
- In other words, it takes **several million electron volts to pry a nucleon** from a typical nucleus, **as compared to just 13.6 eV to ionize an electron** in the ground state of hydrogen.
- This is why **nuclear force is referred to as the "strong" nuclear force**.

Graph of Binding Energy per Nucleon



- The graph rises at low A , peaks very near iron (Fe, $A = 56$), and then tapers off at high A .
- The peak value suggests that the iron nucleus is the most stable nucleus in nature (it is also why nuclear fusion in the cores of stars ends with Fe).
- The reason the graph rises and tapers off has to do with **competing forces in the nucleus**.

Graph of Binding Energy per Nucleon



- At **low values of A** , **attractive nuclear forces** between nucleons **dominate over repulsive electrostatic forces** between protons.
- But at **high values of A** , **repulsive electrostatic forces** between forces **begin to dominate**, and **these forces tend to break apart the nucleus** rather than hold it together.

NUCLEAR FORCE

Why a new force is needed?

- Binding energy per nucleon ≈ 8 MeV, much larger than atomic binding energies (eV range).
- Coulomb repulsion between protons cannot be balanced by gravity (too weak).
- A new strong attractive force is required to hold nucleons together inside nucleus.

CHARACTERISTICS OF NUCLEAR FORCE

Main Properties:

- **Very Strong:** Stronger than both Coulomb and gravitational forces.
- **Short Range:** Effective only up to a few femtometres (fm); falls rapidly beyond.
- **Saturation Property:** Each nucleon interacts only with its nearest neighbours.
- **Charge Independent:** Same for p-p, n-n, and p-n interactions.
- **Attractive–Repulsive Nature:** Attractive beyond 0.8 fm; strongly repulsive below 0.8 fm.
- **No simple formula:** Unlike Coulomb's or Newton's law, its exact mathematical form is unknown.

POTENTIAL ENERGY CURVE

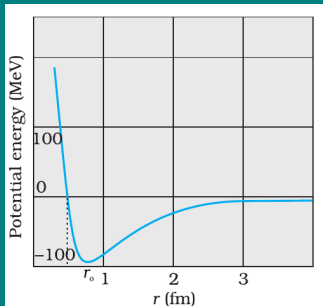


FIGURE 13.2 Potential energy of a pair of nucleons as a function of their separation.

For a separation greater than r_0 , the force is attractive and for separations less than r_0 , the force is strongly repulsive.

Nuclear Potential Energy vs Distance

- Potential energy minimum at ≈ 0.8 fm $\Rightarrow$ stable bound state.
- Attractive for $r > 0.8$ fm.
- Strongly repulsive for $r < 0.8$ fm.